2. Code Architecture Breakdown

This script provides your team with the data substrate needed to demonstrate your governance model on Sunday. Here is how the logic maps directly back to your PRD requirements:

A. The "Frustrated_Masked" Data Paradox (Beating the Face Model)

Look closely at the `Frustrated_Masked` generation block:

- The script generates a `facial_valence_snapshot` between `0.4` and `0.8`. To a standard context-blind AI, **this person registers as happy**.
- However, your rolling state vector captures an `ambient_db` of up to `85.0`, a somatic trigger of `clenched_fists_count > 0`, and `hours_since_break > 3.5`.
- **The Breakdown for Sunday:** When you train a classifier (like a Decision Tree or Random Forest) on this mock data, you can demonstrate to the judges that a model using only the face column gets this wrong, while a model utilizing the full state vector correctly catches the masking dilemma.

B. The "Focused" Data Paradox (Curing Context Blindness)

Look at the `Focused` generation block:

- The `facial_valence_snapshot` is negative (`-0.4` to `0.0`). Standard facial recognition models often misclassify this intense, brow-furrowed look as **angry or hostile**.
- Your pipeline corrects this because the rolling environment notes that the room is at a peaceful focus hum (`40-55 dB`), lights are bright, and breaks are stable.
- **The Breakdown for Sunday:** This directly proves your statement that context clears up "resting hacker face" misclassifications.

C. Structuring the XAI Audit Trail Output

Using this mock data schema, your final presentation's text explanation box can easily query the input rows to parse out *why* it overrode the facial data:

python

```
# Conceptual pseudocode for your Sunday display engine:
if predicted_state == "Frustrated_Masked" and face_prediction == "Happy":
    explanation = (
```

```
        f"The system detected a surface facial smile, but overrode the
classification "
        f"due to environmental provenance. High stress indicators found:
Ambient volume is "
        f"{row['ambient_db']}dB and time since rest is
{row['hours_since_break']} hours."

    )
```